5. Verify Euler's formula for the table given below.

S.No.	Faces	Vertices	Edges
(i)	4	4	6
(ii)	10	6	12
(iii)	12	20	30
(iv)	20	13	30
(v)	32	60	90

Exercise 2.4

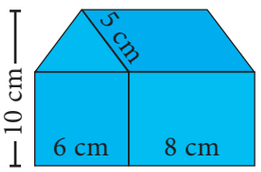
Miscellaneous Practice Problems

1. Two gates are fitted at the entrance of a library. To open the gates easily, a wheel is fixed at 6 feet distance from the wall to which the gate is fixed. If one of the gates is opened to 90°, find the distance moved by the wheel $(\pi = 3.14)$.



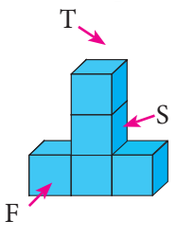
2. With his usual speed, if a person covers a circular track of radius 150 *m* in 9 minutes, find the distance that he covers in 3 minutes $(\pi = 3.14)$.

3. Find the area of the house drawing given in the figure.

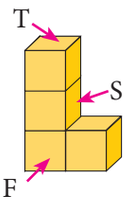


4. Draw the top, front and side view of the following solid shapes

(i)



(ii)

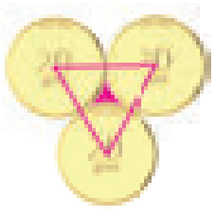


Challenging problems

5. Guna has fixed a single door of width 3 *feet* in his room where as Nathan has fixed a double door, each of width $1\frac{1}{2}$ *feet* in his room. From the closed position, if each of the single and double doors can open up to 120°, whose door takes a minimum area?

6. In a rectangular field which measures 15 $m \times 8m$, cows are tied with a rope of length $3m$ at four corners of the field and also at the centre. Find the area of the field where no cows can graze. $(\pi = 3.14)$

7. Three identical coins each of diameter 6 *cm* are placed as shown. Find the area of the shaded region between the coins. $(\pi = 3.14)$ $(\sqrt{3} = 1.732)$



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